MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG4DSCECT200 PYTHON PROGRAMMING

Duration: 1.5 hours **Maximum Marks: 50**

Students should attempt at least one question from each course outcome to enhance their overall *outcome attainability.*

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries 1 mark

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1	Which of the following is a valid Python identifier? A. 2value B. total-marks C. _count D. class	K	1
2	Which of the following correctly lists Python keywords only? A. if, else, print B. True, False, None C. input, output, return D. int, float, string	U	1
3	Which feature of PyCharm IDE helps in detecting errors while typing code? A. Debugger B. Syntax highlighting C. Version control D. Code deployment	U	1
4	Which control flow statement is used to execute a block repeatedly until a condition becomes false? A. if B. for C. while D. break	K	2
5	Which function is used to terminate the execution of a Python program immediately? A. stop() B. break() C. exit() D. quitloop()	K	2
6	Which of the following functions applies a given function to each item of an iterable? A. filter() B. zip() C. map() D. range()	U	2

7	Which type of chart is most suitable for showing trends over time? A. Bar chart B. Pie chart C. Line graph D. Histogram	U	3
8	Which block in exception handling is always executed, regardless of whether an exception occurs or not? A. try B. except C. else D. finally	U	3
9	Which method of the re module checks for a match only at the beginning of a string? A. find() B. search() C. match() D. split()	A	3
10	Which chart should be used to display the distribution of numerical data? A. Pie chart B. Line graph C. Histogram D. Scatter plot	A	3

1 * 10 = 10

Part B

Short Answer Questions

Answer any **4** Questions

Each question carries 5 marks

11	Explain different Python IDEs, and discuss the features of PyCharm	U	1
12	Explain the types of control flow statements in Python. Distinguish between sequential, decision, and loop control flow statements	K	2
13	Explain the use of break and continue statements in Python with suitable example	U	2
14	Discuss the working of if–else and if–elif–else statements in Python with sample programs	U	2
15	Explain the role of Python libraries in data visualization. Mention any two libraries used for data visualization and their purpose.	U	3
16	Describe how to create a basic GUI window using tkinter. Mention any four commonly used widgets	U	3

4 * 5 = 20

Part C

Essay Questions

Answer any **2** Questions

Each question carries 10marks

17	Explain the basic elements of Python. Discuss data types and type conversions with suitable examples.	U	1
18	Explain operators and expressions in Python, also explain strings and string operations with sample program	U	1
19	Explain looping concepts in Python. Discuss while loop and for loop, including the use of break, continue, range() and exit() functions	U	2
20	Explain exception handling in Python. Discuss raising exceptions using raise, built-in exceptions, and user-defined exceptions	A	3

2 * 10 = 20